

# Ritabrata Chakraborty

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## EDUCATION

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### University of Pennsylvania

*M.S.E. in Robotics, GPA: -/4*

Pennsylvania, USA

*Aug '26 – May '28*

### Birla Institute of Technology and Science, Pilani (BITS Pilani)

*B.E. in Mechanical Engineering — Minor in Data Science, CGPA: 8.57/10*

Rajasthan, India

*Oct '22 – May '26*

## PUBLICATIONS & PEER REVIEWS

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### Journal Publications

- R. Chakraborty and K. Kishore, “*Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation.*” Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026. [🔗](#)
- R. Chakraborty and K. Kishore, “*Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments.*” Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.

### Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, R. Chakraborty, and G. Sartoretti, “*CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction.*” Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**. [🔗](#)
- R. Chakraborty, T. Mian, and P. Kundu, “*An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery.*” Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**. [🔗](#)
- K. Kishore and R. Chakraborty, “*Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring.*” Publication awaiting intellectual property clearance, 2025. [🔗](#)

### Poster Presentations

- R. Chakraborty, “*High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun.*” Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, Visakhapatnam, India, Sep 2024.

### Peer Reviews

- “*GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments.*” Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- “*Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD).*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- “*A Research Paper on IoT-Based Robot for Agriculture.*” Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- Selected as a peer reviewer for **IEEE Gujarat Section Conference**, 2026.

## RESEARCH EXPERIENCE

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### Research Intern

Supervisor: [Dr. Avinash Gautam](#), Associate Professor

BITS Pilani

*May '25 – Aug '25*

- Literature study of vision-language grounded multi-robot coordination and VLMs-based navigation.

### Research Intern

Supervisor: [Dr. Kaushal Kishore](#), Principal Scientist

CSIR-CEERI

*Jan '24 – May '26*

- Monocular vision-based UAV navigation for orchard monitoring using YOLOv11. [🔗](#)
- Multi-critic decomposition for off-policy reinforcement learning (TD3, SAC, DDPG) for autonomous indoor drone navigation. [🔗](#)

## Research Intern

Supervisor: [Dr. Guillaume Sartoretti](#), Assistant Professor

MARMoT Lab, NUS

Mar '25 – Dec '25

- Uncertainty-guided path planning via conditional generative layout prediction (CogniPlan). [🔗](#)
- Vision-attention-driven autonomous navigation with semantic understanding, extending [CogniPlan](#) and [ViNT](#).

## Research Assistant

Supervisor: [Dr. Mani Shankar Dasgupta](#), Associate Professor

BITS Pilani

Apr '24 – Oct '25

- Physics-based launcher mechanical design and FEM analysis. [🔗](#)
- Computer vision pipeline for 3D trajectory reconstruction and shot classification using [PnLCalib](#).
- Finite element analysis of running track surfaces using hyperelastic material models.

## Research Intern

Supervisor: [Dr. Pradeep Kundu](#), Assistant Professor

KU Leuven

Sep '24 – May '26

- Synthetic data generation for bearing fault diagnosis under data scarcity. [🔗](#)
- Physics-informed generative modeling for bearing fault diagnosis under data scarcity, validated on [LiteFormer](#).

## Research Assistant

Supervisor: [Dr. Rajneesh Choubisa](#)

BITS Pilani

Aug '24 – Dec '24

- Primal world beliefs study surveying engineering undergraduates on core worldview dimensions. [🔗](#)

## Research Intern

Supervisor: [Dr. Rishi Verma](#), Head (ESDS), PPEMD

BARC

Jun '24 – Jul '24

- Image-based visual servoing for automated radar control and UAV tracking using YOLOv8 and DeepSORT.
- Impact testing and analysis of metal-composite shields using electromagnetic railguns.

## INDUSTRY EXPERIENCE

### ML Intern

Supervisors: [Mr. Siddarth Malreddy](#), Tech Lead Manager & [Mr. Ishan Nigam](#), Senior ML Engineer

Uber

Jul '25 – Dec '25

- Object annotation automation and anomaly detection in [uLabel](#).

### Subject Matter Expert

Self-supervised

Knowledge Cultivator

Nov '23 – Feb '24

- Technical problem-solving and academic content authoring using MATLAB, CAD, and Python.

## PROJECTS

### Vision-Guided Robotic Arm for Simulated Oral Cancer Screening

Aug '25 – Dec '25

### Autonomous Drone Navigation | MathWorks Global Student Drone Challenge 2025

Mar '25 – Apr '25

### Air Quality Index Forecasting Using Time Series Models [🔗](#)

Nov '24 – Nov '24

### 3D Lunar Surface Modeling from DEM Data | ISRO Immersion Challenge 2024 [🔗](#)

Jul '24 – Jul '24

### Cricket Team Optimization | AMEX Campus Challenge 2024 (Product Track) [🔗](#)

Jun '24 – Jul '24

### ExoMy Rover Navigation and UR3 Arm Motion Planning | ERC 2023 Remote [🔗](#)

Apr '23 – Sep '23

## TEACHING EXPERIENCE

### Teaching Assistant

CIS 4190/5190: Applied Machine Learning, University of Pennsylvania

Aug '26 – Dec '26

CS F320: Foundations of Data Science, BITS Pilani

Jan '26 – May '26

ME F218: Advanced Mechanics of Solids, BITS Pilani

Jan '25 – May '25

ME F216: Materials Science and Engineering, BITS Pilani

Sep '24 – Dec '24

## HONOURS & SCHOLARSHIPS

Institute Merit-cum-Need Scholarship — Consecutive Recipient (Top 8% of Students) Oct '22 – May '26

## AWARDS & ACHIEVEMENTS

Finalist — Meta PyTorch OpenEnv Hackathon (India) [🔗](#)

Apr '26

3<sup>rd</sup> Place — MathWorks Global Drone Student Challenge 2025

Mar '25

Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) [🔗](#)

Jul '24

Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 [🔗](#)

Jul '24

5<sup>th</sup> Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals)

Sep '23

LEADERSHIP EXPERIENCE

|                                                                                                                                                       |                          |
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| <b>Deputy Director of Academic Programming</b><br><i>Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania</i>               | <i>Aug '26 – Present</i> |
| <b>Vice-President</b><br><i>South Asian Association (Rangoli), University of Pennsylvania</i>                                                         | <i>Aug '26 – Present</i> |
| <b>President &amp; Secretary</b><br><i>Mechanical Engineering Association (MEA), BITS Pilani</i>                                                      | <i>Jun '24 – May '26</i> |
| <b>President &amp; Tech Fest Coordinator</b><br><i>Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani</i> | <i>Oct '24 – Jul '25</i> |
| <b>Project Manager</b><br><i>Tinkerer's Lab (TL), BITS Pilani</i>                                                                                     | <i>May '24 – Jul '25</i> |
| <b>Treasurer</b><br><i>Bengali Cultural Association (Moruchhaya), BITS Pilani</i>                                                                     | <i>May '24 – Jul '25</i> |

VOLUNTEERING EXPERIENCE

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|-------------------------------------------------------------------------------|--------------------------|
| <b>Volunteer</b><br><i>Junoon, National Service Scheme (NSS), BITS Pilani</i> | <i>Sep '23 – Sep '23</i> |
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PROFESSIONAL ACTIVITIES

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|---------------------------------------|-------------|
| <b>Graduate Student Member — IEEE</b> | <i>2026</i> |
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COMPETITIVE EXAMINATIONS

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|-------------------------------------------------------------------------------|-------------|
| <b>GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3)</b>                      | <i>2025</i> |
| <b>TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28)</b> | <i>2025</i> |
| <b>JEE Advanced: All India Rank 14372</b>                                     | <i>2022</i> |

TECHNICAL SKILLS & COURSEWORK

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| <b>Relevant Coursework:</b> Machine Learning, Deep Learning, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Computer Programming, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins |
| <b>Programming Languages:</b> Python, C++, C, C#, Java, Bash (Shell)                                                                                                                                                                                     |
| <b>Robotics &amp; Simulation:</b> ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS                                                                                                        |
| <b>AI &amp; Data Science:</b> PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Excel                                                         |
| <b>Cloud &amp; MLOps:</b> AWS (S3, EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), Docker, Git, GitHub Actions                                                                                                                            |
| <b>Hardware &amp; Embedded Systems:</b> NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR                                                                                                                                 |
| <b>CAD &amp; Mechanical Simulation:</b> ANSYS Mechanical, SolidWorks, Fusion 360                                                                                                                                                                         |
| <b>Leadership &amp; Communication:</b> Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing                                                           |
| <b>Productivity Software:</b> Microsoft Office (Excel, PowerPoint, Word), Google Workspace                                                                                                                                                               |